Imitation-Reinforcement Learning for Dynamic Multi-Period Mean-Variance Portfolio Optimization

Abstract

The research focuses on a novel hybrid imitation-reinforcement learning (IL-RL) framework that trains a diffusion-policy agent to replicate and surpass the dynamic multi-period Markowitz optimal portfolios in a controlled yet realistic financial environment. Daily asset returns are simulated from a multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ with stochastic regime shifts, while the expert demonstrator solves a rolling-horizon quadratic program that explicitly incorporates transaction costs and leverage constraints.

The agent is first warm-started via advanced imitation learning (DAGGER + GAIL with Wasserstein regularization) using a transformer-diffusion architecture, then fine-tuned with proximal policy optimization (PPO) under an economically meaningful reward that penalizes deviation from the expert while maximizing certainty-equivalent utility.

Theoretical contributions include sublinear regret bounds on imitation error under covariance misspecification and regime switching. Extensive Monte-Carlo experiments (500 independent runs) demonstrate superior out-of-sample Sharpe ratios, lower turnover, reduced utility loss, and orders-of-magnitude faster inference compared with rolling-window Markowitz, pure RL baselines, and state-of-the-art portfolio optimization methods. This work bridges model-based portfolio theory with modern generative IL-RL, offering a scalable, robust, and theoretically grounded solution for high-dimensional dynamic asset allocation.

1 Project Overview and Objectives

This advanced project elevates classic Markowitz replication into a theoretically rigorous, multi-period, friction-aware framework. The goal is to develop an AI policy that not only mimics but also improves upon the expert under realistic market frictions.

Key factors of the project:

- Stochastic regime shifts and transaction costs in the simulation.
- Multi-period rolling-horizon expert demonstrator.

- Diffusion-based generative policy with transformer state encoder.
- Hybrid IL-to-RL training pipeline.
- Sublinear regret analysis and economic significance metrics.

The entire pipeline remains fully simulation-based and reproducible.

2 Required Tools and Libraries

Python 3.10+ with:

- `numpy`, `pandas`, `scipy.optimize`, `cvxpy` — simulation and expert QP.
- `torch`, `diffusers` (or custom diffusion policy), `transformers` — policy network.
- `stable-baselines3` — PPO fine-tuning.
- `scikit-learn`, `statsmodels` — metrics and statistical tests.
- `matplotlib`, `seaborn` — visualizations.

3 Implementation Steps

3.1 Step 1: Advanced Stock Return Simulation with Regime Shifts

1. Set N assets (50–200), horizon $T = 10{,}000$.
2. Define base $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$; introduce $K = 3$ latent regimes with periodic or Markov-switching transitions.
3. At each t, draw $\mathbf{r}_t \sim \mathcal{N}(\boldsymbol{\mu}_{r_t}, \boldsymbol{\Sigma}_{r_t})$ where regime r_t evolves according to a transition matrix.
4. Incorporate proportional transaction costs $c \in [0.001, 0.005]$.

3.2 Step 2: Multi-Period Markowitz Expert Demonstrator

Solve a rolling-horizon quadratic program at each rebalance:

$$\boldsymbol{w}_t^* = \arg\max_{\boldsymbol{w}} \mathbb{E}_t \left[\sum_{\tau=t}^{t+H} \boldsymbol{w}_\tau^\top \boldsymbol{\mu}_\tau - \frac{\gamma}{2} \boldsymbol{w}_\tau^\top \boldsymbol{\Sigma}_\tau \boldsymbol{w}_\tau - c\|\Delta \boldsymbol{w}_\tau\|_1 \right]$$

subject to budget, no-shorting, and leverage constraints (H = horizon, e.g., 21 days). Use CVXPY for efficient solving.

3.3 Step 3: State-Action Representation

State s_t: transformer-encoded window of returns, regime features (via learned embedding), estimated moments, and cost-adjusted history. Action a_t: portfolio weights $\boldsymbol{w}_t \in \Delta^{N-1}$.

3.4 Step 4: Hybrid IL-RL Training Pipeline

1. **Phase 1 (Imitation Warm-Start)**: Train diffusion policy via DAGGER + GAIL with Wasserstein-1 regularization on expert trajectories.
2. **Phase 2 (RL Fine-Tuning)**: Use PPO with reward $r_t = \text{CE utility} - \lambda\|\boldsymbol{w}_t - \boldsymbol{w}_t^*\|_2$, where CE is certainty-equivalent return.
3. Policy: Transformer encoder + conditional diffusion model (noise schedule tailored to simplex constraint via projected denoising).

3.5 Step 5: Theoretical Analysis

Derive regret bound:

$$\text{Regret}(T) \leq \mathcal{O}\left(\sqrt{T \log N} + \epsilon_{\text{est}} T\right)$$

where ϵ_{est} is covariance estimation error, proven via online learning theory and Wasserstein stability of the diffusion policy.

3.6 Step 6: Rigorous Evaluation

500 Monte-Carlo runs. Metrics:

- Annualized Sharpe ratio, Sortino ratio, certainty-equivalent utility.
- Turnover, maximum drawdown, economic significance (utility loss vs. expert).
- Inference latency, zero-shot generalization to unseen regimes/covariances.
- Ablation studies (pure IL vs. hybrid, diffusion vs. deterministic policy).

Statistical tests: Diebold-Mariano, Giacomini-White, and bootstrap confidence intervals.